BOUNDED STUDY INDEX | XR-ACCEL-01 | REFERENCE: SCALAR RV64

CLAIM UNDER REVIEW The accelerator-memory organization improves XR efficiency over scalar and vector-capable RISC-V alternatives within 3 W, preserving QoS and deployability.

Intent

Improve mobile XR efficiency without violating power, QoS, memory, or deployment constraints.

Task

Compare scalar RV64 (reference), vector-capable, and accelerator-memory on the same XRBench slice.

Design space

Legal accelerator, memory, and vector choices; software redesign and full SoC changes are out.

Representation

XRBench traces, architecture parameters, compiler assumptions, power model, and uncertainty record.

Environment

Simulator or cost model plus workload harness; the contract defines actions and observations.

Method role

Generate candidates, predict cost, optimize, challenge assumptions, and explain mechanisms.

Feedback budget

Many cheap proxies, fewer simulations, and two high-fidelity checks before claim review.

Evidence

Links to runs, three-way comparisons, sensitivities, provenance, and limits in the evidence record.

Failed / rejected

Links to power violations, QoS misses, bandwidth overruns, proxy mismatches, and discarded options.

Rejection checks + authority

Power, QoS, proxy, and cross-tool checks may block or weaken the claim; the architect sets disposition.

Commitment boundary

Authorize a higher-fidelity study only; no RTL, integration, or product commitment.

Accountable decision

The named architect owns scope changes, waivers, claim strength, escalation, and commitment.

The card indexes the study, claim, artifacts, evidence, and decision. It does not replace the records.